HOLLOWHAVEN

Technical Manual

SYSTEM DOCUMENTATION

2026.11 EDITION 01

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Version: 2026-11-06. English translation of the revised Chinese edition; technical baseline: Git e40f8b5. The manual covers the system, technology stack, hardware, software, structure, assembly, testing and replication. Missing dimensions and measured results remain explicitly pending.

1. Project introduction and technical objectives

HollowHaven is a course prototype developed for an artificial-hollow shelter concept for Greater Gliders. It explores the integration of environmental monitoring, automatic drip watering and visitor-event recording in one device. An ESP32 collects environmental and equipment data, operates a servo-controlled gravity-fed water supply, and sends data to a computer dashboard for observation and recording.

The enclosure combines Blender modelling with layered assembly: a perforated structural layer, a capillary water-distribution layer and an outer bark layer. The structure provides mounting locations for the entrance, sensors and water system; the absorbent material distributes water and the bark forms the exterior covering.

1.1 Three main functional chains

Function	Input and processing	Output
Environmental monitoring	Temperature, humidity, light, water level and two material-moisture measurements	Dashboard status, trends, CSV records and maintenance alerts
Automatic drip watering	Valid light and temperature conditions start the servo valve; temperature is checked periodically	Gravity-fed watering and command status; moisture probes help check water-delivery response
Visitor-event recording	Consecutive distance readings confirm a possible entry	Approximately 30 seconds of recording, WAV storage and optional user-initiated audio analysis

1.2 Technical objectives and validation scope

The technical objective is to connect sensor acquisition, local device control, LAN transmission, computer storage and browser display, and integrate the electronics into the layered enclosure. Evaluation should establish whether these stages work together consistently, repeatably and maintainably.

The available materials support descriptions of implemented code and assembly relationships shown in photographs. They do not establish a verified temperature reduction, long-term waterproofing or suitability for wildlife. Distance events and audio analysis provide observational clues rather than species identification.

1.3 Expected reader outcomes

Readers should be able to explain each technical layer, assemble a bench prototype using the wiring table, deploy the specified software version, check basic functions and identify the dimensional and fabrication information needed for complete mechanical replication.

2. System architecture and technology stack

2.1 System responsibilities

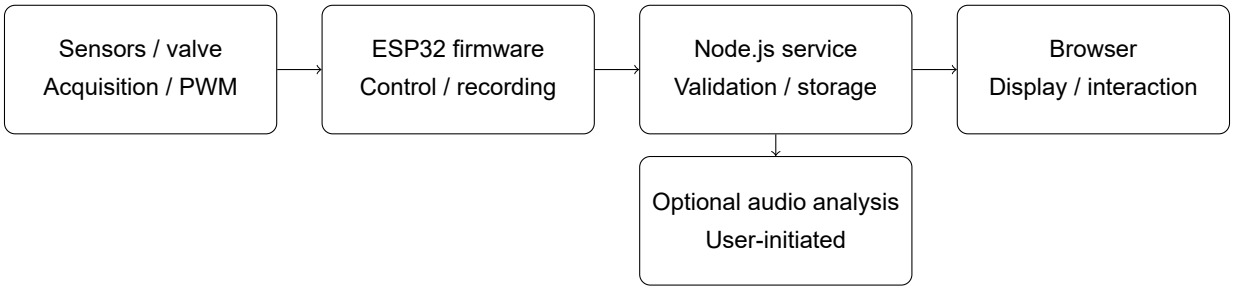


Figure 1. System architecture. Boxes follow the data-processing sequence. Valve control runs on the device; optional audio analysis is initiated by the user.

Layer	Current technology	Responsibility
Hardware	ESP32-WROOM-32, sensors, MG90S servo	Physical measurements and valve actuation
Firmware	C, ESP-IDF, PlatformIO	I²C / ADC acquisition, PWM, state machines and audio buffering
Communication	Wi-Fi LAN, UDP, TCP	Discovery, telemetry and audio transfer
Local service	Node.js	Device connections, protocol validation, file storage and web interfaces
Browser	HTML, CSS, JavaScript	Live status, trends, recordings and settings
Data	CSV, WAV, JSON	Telemetry, audio, analysis results and non-sensitive settings
Structural design	Blender and layered enclosure assembly	Curved shell, ventilation, water distribution and exterior covering

2.2 Architectural choices and operating boundaries

Local device control makes valve decisions independent of whether the dashboard is open. The computer receives, stores and displays data. Disconnecting it interrupts reception but does not stop device-side control. The system cannot recover a complete history of telemetry collected while offline.

Telemetry and audio use separate TCP connections to handle status data and audio files. The browser accesses the local Node.js service rather than reading sensors directly. Optional audio analysis is proxied by the service, and its results do not influence valve decisions.

3. Hardware design and implementation

3.1 Components and purposes

Component	Quantity	Interface and purpose
ESP32-WROOM-32 development board and matching expansion board	One each	Acquisition, control and Wi-Fi
SHT3x temperature/humidity module	One	First I²C bus; cavity-air measurements
VL53L0X distance module	One	Second I²C bus; entrance-area distance
Capacitive moisture module	Two	ADC; wetting response at different material locations
Analogue light and water-level modules	One each	ADC; opening condition and refill alert
SEN0487 microphone	One	Analogue audio and sound statistics
MG90S servo and valve mechanism	One each	PWM-driven mechanical valve
Tank, tubing, capillary matting and fasteners	As required by the structure	Gravity delivery, distribution and mounting

Confirm the supply and output levels of complete modules before purchase; a bare chip cannot directly replace a module. Exact purchased light, water-level and capacitive-moisture module models, valve and connector specifications remain to be documented. Tools should include a USB data cable, multimeter, small screwdriver, insulation materials and cable ties.

3.2 Power and interface design

Power the ESP32 through its onboard USB connector from a regulated 5V USB source. Use 3.3V-compatible sensor modules. Supply the MG90S from a separate regulated 5V branch with a common ground to the ESP32, not from 3V3. Validate the supply against the actual servo load and voltage drop during movement.

Keep all analogue inputs within 0–3.3V and use AO outputs. Analogue acquisition uses ADC1 to avoid the classic ESP32's ADC2 resource conflict during Wi-Fi operation. Temperature/humidity and distance modules use separate I²C buses, which should be checked independently during troubleshooting.

DS18B20 and the external water-level-module power-switch circuit are currently disabled. Definitions retained in a configuration file do not mean these functions should be connected or enabled.

3.3 Hardware diagram and connection table

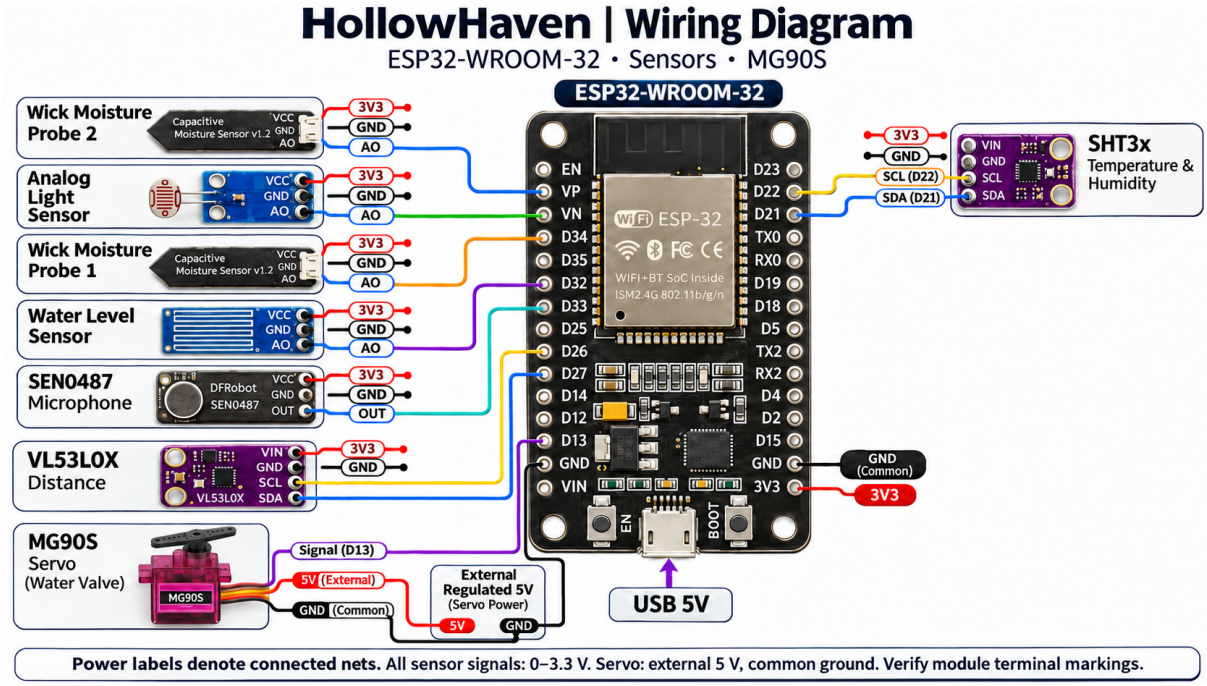


Figure 2. Wiring illustration from the original manual. Colours aid identification; connect by module terminal labels and the table below.

Module terminal	ESP32 / supply terminal	Check
SHT3x SDA / SCL	GPIO21 / GPIO22	First I²C bus
VL53L0X SDA / SCL	GPIO27 / GPIO26	Second I²C bus
Moisture 1 AO	GPIO34	First wetting point
Moisture 2 AO	GPIO36 / VP	Not GPIO35 or VN
Light AO	GPIO39 / VN	Do not use DO
Water-level AO	GPIO32	Raw ADC, not millilitres
SEN0487 A	GPIO33	Analogue audio
Sensor VCC / VIN and GND	3V3 and common GND	Confirm 3.3V compatibility
MG90S signal, positive and negative supply	GPIO13, separate 5V, common GND	Check manufacturer terminal markings

VP means GPIO36 and VN means GPIO39; neither denotes supply polarity. D-prefixed board labels refer to GPIO numbers, whereas computer COM numbers identify programming connections. Check the actual expansion board's S/V/G order, using continuity tests with power off if necessary. Do not infer hidden connections from photographs.

3.4 Sensor and water-system principles

SHT3x measures cavity temperature and humidity; temperature participates in current watering control. Light, water level and material moisture are used as raw ADC readings, not calibrated lux, millilitres or moisture percentage. Two

moisture locations indicate whether water reaches different points, but cannot establish uniform wetting across the entire surface.

VL53L0X measures entrance-area distance, and software uses consecutive readings to confirm possible entry and exit. SEN0487 supplies analogue sound for recording after an entry event. Keep the microphone suitably separated from water and servo vibration.

The water route is elevated tank, servo valve, tubing and external distribution layer. Gravity delivers water and capillary action spreads it without an additional pump. Evaporation depends on temperature, humidity, airflow, supply rate and material contact; evaluate its effects through comparative measurements.

3.5 Mechanical movement and feedback boundaries

PWM commands drive the servo and its mechanical coupling to the valve. The command changes gradually between closed and open endpoints, taking approximately 13 seconds. Dashboard state reflects commands; there is no flow meter or valve-position feedback. Check actual discharge and material wetting to establish the physical response.

Initially disconnect the linkage, check direction and endpoints, and then reconnect it. Replacing the servo, valve or bracket requires recalibrating pulse endpoints and mechanical limits to avoid stalls. Tank height, tube diameter and valve design influence flow and should be recorded with assembly details.

3.6 Replication checks for this chapter

Verify every signal, supply and ground connection. Keep analogue output within input limits and check for ESP32 resets during servo operation. Validate without water before checking the filled water path. The temperature/humidity probe measures cavity air, whereas moisture probes measure the distribution material; their mounting purposes must remain distinct.

4. Software design and implementation

4.1 Modules and responsibilities

Module / location	Main responsibility
project/firmware/main/	Startup, sensor acquisition, networking and task organisation
sensors.c, analog.c	Digital sensors, analogue values and sound acquisition
presence_sensor.c, shared/presence.h	Distance-event confirmation and entry/exit states
servo.c, shared/valve_control.h	PWM movement and watering-control state
audio_record.c	Audio buffering, recording and transfer
website/protocol.js	Telemetry parsing and audio validation
website/server.js	Connections, storage, web interfaces and analysis proxy
website/public/	Pages, styling, interactions and control-state display

The device periodically samples sensors, maintains validity flags and control state, and transmits telemetry. Values, validity and freshness must be distinguished: invalid or stale data cannot be treated as normal measurements. Software cannot necessarily identify floating ADC readings from disconnected modules.

4.2 Drip-watering control logic

While closed, opening is requested only when light data is valid and ADC < 500, and temperature is valid and above 30°C. Timing starts when the servo reaches its commanded fully open state. Hold for at least 60 seconds, then check temperature every minute: close only if it is valid and below 26°C; otherwise remain open until the next check.

Changes in light after opening do not terminate the current cycle. Invalid temperature does not trigger closure. Low water level raises an alert but currently does not interlock valve opening. Test each of these implemented behaviours separately.

Opening conditions → gradual servo movement → commanded fully open → hold 60 seconds → temperature check → close or continue waiting.

4.3 Current parameters and calibration locations

Item	Current value / behaviour	Replication consideration
Opening	Valid light ADC < 500 and temperature > 30°C	Check ADC direction under shaded / illuminated conditions
Closing	Check every minute after at least 60 seconds fully open; temperature < 26°C	Equality does not satisfy a strict comparison
Servo PWM	50Hz; open 1200µs, closed 2500µs	Calibrate to the valve; these are not angles
Servo ramp	10µs every 100ms	Approximately 13 seconds full travel
Material moisture	ADC < 2000 wet; ≥2300 dry	Retain prior state in between; initial or invalid state is unknown
Water-delivery alert	Fully open ≥60 seconds and either point still dry	Possible fault only; inspect water path manually
Low-water alert	ADC < 50	The “below 50ml” label is not volumetric calibration
Entry / exit	Three consecutive readings ≤189mm / five ≥220mm	Approximately 100ms sampling; remeasure background after enclosure installation
Event recording	Approximately 30 seconds	Continuous occupancy does not repeatedly trigger

Pins, feature switches and some thresholds are in project_config.h. Watering logic is in shared/valve_control.h; material moisture, maintenance alerts and entry/exit states are in the corresponding shared modules. Threshold changes require editing, rebuilding and flashing firmware and updating calibration records.

4.4 Distance triggers and audio buffering

Consecutive-reading confirmation and different entry/exit thresholds reduce boundary jitter. An entry event requests recording, after which PCM and metadata are sent to the computer. Flash capacity is limited: offline recordings can be overwritten, so the buffer is not an indefinite archive.

Audio transfer includes CRC32. The computer validates the data, writes WAV and acknowledges reception. CRC32 checks transfer integrity, not absence of sampling gaps or the identity of the recorded subject.

4.5 Communication and storage

Link	Address / port	Purpose
Device discovery	UDP 9002	Discover the ESP32 on the LAN
Telemetry	TCP 9000	Version header and CSV rows with sequence, time and status
Audio	TCP 9001	Metadata, PCM and validation
Browser	http://127.0.0.1:8766	Local Node.js service

Connect the computer and ESP32 to the same router. Telemetry includes boot identifier, sequence number, uptime and validity; the computer separately records reception time to distinguish device time from arrival time. If discovery fails, manually enter the IP printed on the serial console.

Storage location	Content
website/data/session_*.csv	Service-session telemetry
website/recordings/	WAV recordings
website/data/results.json	Audio-analysis results
website/data/settings.json	Non-sensitive settings

4.6 Dashboard and optional audio analysis

The dashboard provides live state, trends, maintenance alerts, CSV export, recording search, playback, downloads and WAV imports. Trends use up to 900 recent samples held by the backend. The recording list displays up to 250 recent WAV files; this display limit is not an automatic deletion rule.

After the user selects audio and explicitly starts analysis, Node.js calls the configured external service. Results are candidates requiring verification. API keys remain in service-process memory and must be entered again after restart.

There is no manual valve switch in the dashboard. Closing the page or pausing reception does not change firmware thresholds. Troubleshoot device acquisition, network transfer, service reception and browser display separately.

5. 3D modelling and mechanical design

5.1 Model structure and design intent

Blender is used for structural design. Existing screenshots show a curved shell, triangular openings and continuous edges. Openings provide ventilation paths; edges provide a basis for assembly. Final strength and fastening must be validated against actual wall thickness, material and connection methods.

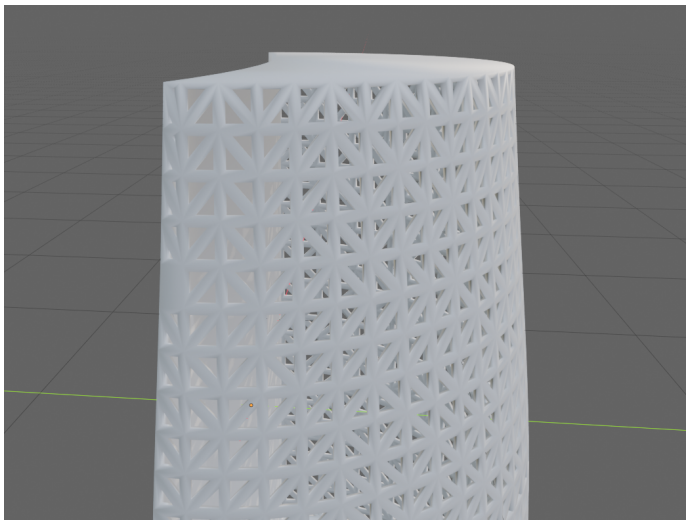


Figure 3. Detail of the ventilated structure: curved mesh and triangular openings.

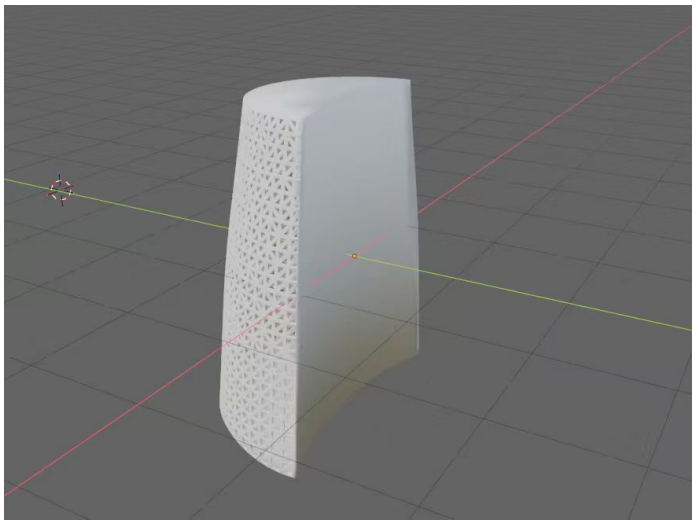


Figure 4. Overall model: curved shell, openings and edges. Neither image provides manufacturing dimensions.

The repository's modeling directory contains Blender assets. Match the chosen file to the final physical revision before checking units, scale and exported meshes. Image proportions do not replace dimensioned drawings.

5.2 Three-layer structure and installation layout

From inside out, the enclosure consists of a ventilated structural layer, capillary water-distribution layer and bark layer. Continuous material contact spreads water. Keep the wet layer and drainage away from the resting space and electronics. Preserve the entrance and necessary ventilation when fastening the exterior.

Position the temperature/humidity probe to represent cavity air. Keep the distance sensor's field of view clear of wires and shell surfaces, and keep the microphone away from water and servo vibration. Place moisture probes at

different distribution points. Mark the tank, valve, service opening and wiring attachments on the final structural drawing.

5.3 Parameters required before manufacture

Object	Measurements or decisions still required
Body and lid	Height, width, depth, wall thickness, material, split and joining method
Entrance and interior	Entrance diameter, height above base and usable cavity dimensions
Ventilation openings	Diameter or side length, count, spacing and covered area
Tank and water path	Capacity, relative height, tube diameter, valve and drainage route
Servo and electronics compartment	Bracket dimensions, hole locations, limits and access
Layer materials	Mat thickness and layout dimensions; bark attachment
Fabrication and printing	Equipment, material, layer height, infill, supports and orientation

Complete values were not supplied, so these materials cannot produce accurate manufacturing drawings. The original manual’s future full-scale benchmark is not the measured size of this prototype. Replicate from confirmed models and measurement tables.

5.4 Modelling and export advice

Determine actual component and water-system clearances before adjusting the exterior. Check units, mesh closure, wall thickness, holes and assembly tolerances. Validate fit with small connection samples before manufacturing the full body. Deliver editable .blend files, manufacturing meshes and dimensioned drawings. Determine print settings experimentally for the actual equipment and material rather than inferring them from screenshots.

6. Enclosure fabrication and system assembly

6.1 Physical assembly references

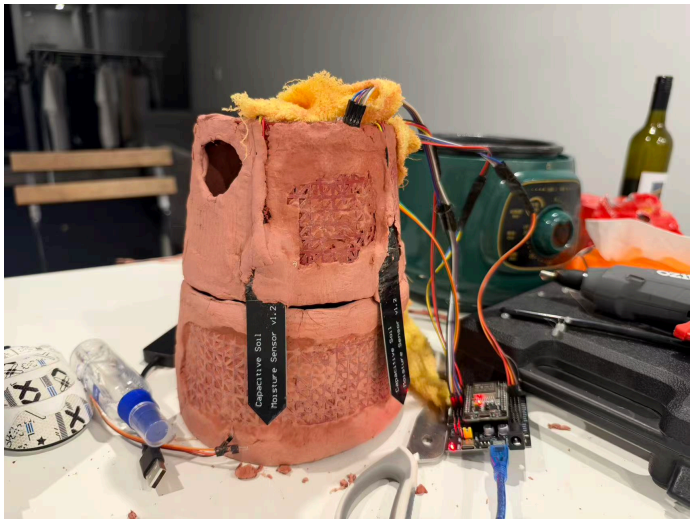


Figure 5. Assembly stage showing the body, entrance, two capacitive moisture probes and externally connected electronics.

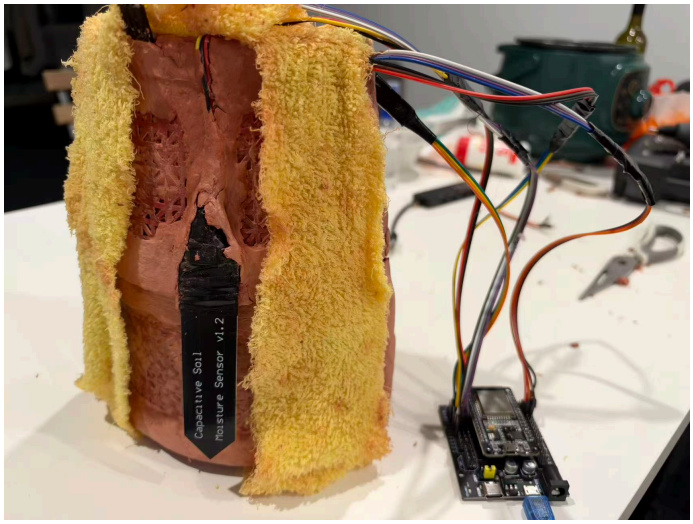


Figure 6. Capillary material installation: maintain contact with the structure and check wetting at different locations.

Photographs illustrate assembly relationships in the existing prototype. Complete records of material composition, adhesives, fasteners and waterproof rating are not available and should be completed before replication.

6.2 Recommended assembly sequence

Stage	Work	Completion check
1 Bench validation	Connect sensors, firmware and dashboard; initially unload the servo	Valid data and correct movement direction
2 Body preparation	Manufacture, clean edges and check entrance and space	No burrs; components fit and remain accessible
3 Dry-fit water path	Fit tank, tubing, valve and servo bracket	No interference or movement against limits

Stage	Work	Completion check
4 Distribution layer	Install continuous capillary mat and two moisture points	Good contact; ventilation and entrance unobstructed
5 Sensors	Install temperature/humidity, distance and microphone modules	Appropriate positions and clear sensing area
6 Wiring	Secure electronics and organise cables	Separate water and electronics; reliable connections and access
7 Bark covering	Attach bark pieces while retaining entrance	No crushed cables or obstructed water/service paths
8 Wet commissioning	Add a little water, inspect, then run a complete cycle	No leakage; recordable wetting response

6.3 Distribution-layer and enclosure considerations

Capillary matting distributes water according to contact continuity, inclination and supply conditions. Avoid unsupported areas, folds and blocked openings. Check upper, lower and circumferential wetting at the expected installation angle.

Maintain repeatable moisture-probe contact so disassembly does not change readings. Make electronics, power connectors and tubing joints inspectable. Exposed wiring in photographs is not a specification for long-term outdoor installation.

After assembly, record sensor positions, tank height, tubing routes and material specifications, and photograph the interior before covering it to support maintenance and future replication.

7. System testing and result recording

7.1 Available evidence and result boundaries

This revision checks the existing manual, team images, repository configuration and control logic; it does not include new physical experiments. Code supports implementation descriptions and photos support assembly relationships. Neither replaces whole-system acceptance or thermal measurements. The following is a proposed test plan, not a list of passed tests.

Test	Procedure and observation	Record
Power and wiring	Check continuity unpowered, add modules individually and observe servo voltage drop	Voltages, unexpected resets and wiring revision
Sensors	Vary light, wetness, water level and distance; check temperature/humidity validity	Raw ADC, reference conditions and positions

Test	Procedure and observation	Record
Watering control	Cover thresholds, 60-second hold, invalid temperature and low water	Commands, physical motion and timing
Water distribution	Observe discharge and two wetting points; vary inclination	Response times, wetting distribution and leaks
Events and audio	Enter, remain, leave and re-enter	Event count, WAV duration and playback
Network and storage	Save CSV/WAV, disconnect and restart service	File integrity, recovery and gaps
Enclosure integration	Check obstruction, vibration, access and separation of water/electronics	Problem photos and corrective actions

7.2 Evaluating cooling performance

To report cooling, establish comparable watered and unwatered conditions. Record ambient temperature/humidity, irradiation or light conditions, airflow, water supply, sensor locations and measurement times. Preserve raw data, compare cavity-temperature changes under comparable conditions, and document repetitions and sensor error.

If only a single temperature decrease is observed, report its conditions and interval rather than inferring long-term performance. Two wet probes demonstrate local material response, not whole-system cooling or uniform watering.

7.3 Test-record format

For each run record date, operator, firmware commit, component models, calibration values, positions, input, expected behaviour, actual observations, file names and conclusion. Quantitative results remain to be supplied by the team. Mark untested items as pending and retain failures and retest records.

8. Replication advice and future improvements

8.1 Software deployment

Prepare Node.js 18 or later and PlatformIO. Clone and select the version checked for this manual. If later revisions change pins, protocols or thresholds, recheck the associated instructions.

```
git clone \url{https://github.com/SMGG2302/comp-5047}.git
cd comp-5047
git checkout e40f8b5
```

In project/firmware/main, copy wifi_credentials.example.h to wifi_credentials.h and enter the local 2.4GHz router credentials. Exclude real credentials from shared packages. Retain partitions.csv to preserve the audio-buffer layout.

Run these commands in project/firmware. COM7 is an example; substitute the actual port from the device list.

```
pio run
```

```
pio device list
pio run -t upload --upload-port COM7
pio device monitor -p COM7 -b 115200
```

In website, run the service and open the local address in a browser.

```
node server.js
http://127.0.0.1:8766
```

Connect computer and ESP32 to the same router, check client isolation and initially select auto for the device address. If discovery fails, use the serial-console IP and check UDP 9002 and TCP 9000 / 9001 connectivity. Avoid running another program that occupies device receiver connections.

The first build downloads the toolchain. If a Windows path containing Chinese characters causes failure, consult build-ascii.ps1, adapting its machine-specific paths to the actual computer.

8.2 Staged replication and calibration

Proceed through component preparation, bench wiring, firmware flashing, service startup, individual-module validation, enclosure assembly, water-system connection and whole-system testing. Save results at each stage before proceeding to make faults easier to locate.

Record dry, wet and intermediate material ADC values, then verify light direction, water-level threshold and empty-cavity background distance. Calibrate servo endpoints on the actual mechanism. Do not reuse thresholds unchanged after replacing modules or relocating sensors. Include low-water and invalid-temperature behaviours in testing rather than assuming automatic protection exists.

8.3 Common faults

Symptom	First checks
No COM device	USB data capability, driver, cable and port
Reset during servo motion	Supply branch, common ground, voltage drop and stalling
Second moisture channel abnormal	VP / GPIO36 rather than GPIO35 or VN
Invalid I²C readings	Both buses' SDA / SCL, supply and ground
Drifting or extreme analogue readings	AO connection, voltage range, contact and module presence
Dashboard cannot find device	Same LAN, isolation, firewall, IP and receiver connections
Material stays dry after opening	Tank, obstruction, actual valve position, material placement and contact

8.4 Handover information and next improvements

A complete replication package includes version-fixed code, BOM and purchased models, wiring diagram, Blender sources, manufacturing models and dimensioned drawings, print settings, assembly photos, calibration tables and acceptance records. Priorities are completing mechanical dimensions, attachments, material specifications and quantitative tests.

Future work can evaluate low-water interlocks, invalid-temperature policy, flow or valve-position feedback, long-term power, drainage and outdoor protection. These are improvement directions rather than confirmed current capabilities.

8.5 Source basis

This edition is based on SMGG2302/comp-5047 snapshot e40f8b5, with particular checks of project_config.h, shared/valve_control.h and website/package.json. Other implementation descriptions and images are drawn from the repository's Chinese technical manual and its LaTeX asset package. Repository: `https://github.com/SMGG2302/comp-5047`.

The revision adds information structure, technical explanation and replication organisation; proposed tests are not presented as measured results. Update the version and retain a change record when manufacturing parameters and experimental data are supplied.